area, calculate its location and localise its position in the world map. High-definition 3D maps mark road features such as lane centres and markers, curbs, traffic signs and lights, crosswalks and speed bumps, and help in figuring the safest and most efficient route to the destination. Finally, the vehicle's computer signals the relevant hardware systems for steering, acceleration, brake, or taking a turn whenever required. The greater the movement, the smarter the system gets by learning more about the path and behaviour of drivers via the implementation of computer vision and advanced machine learning.

There are no drivers or passengers, and lower speed operation at/or below 40kmph enables more time for reaction and prevents collisions. The vehicle emits zero emissions, and alternative fuels are used to power these vehicles. Also, there are no parking and network connectivity issues for these lightweight and narrow-framed machines. Extensive testing is conducted before the vehicle begins driving on roads, and even during operation, it checks for errors hundreds of times per second for safety and collision avoidance. It never gets distracted like human drivers and focuses on defensive driving.

With multiple compartments, multiple trips can be combined that lower the energy, lessen congestion on roads and economic costs. Customers do not have to bend over to lift items. The doors open without swinging far out, enabling the bot to avoid blocking cars. Unlike traditional vehicles, softer materials are used for the front and rear panels, and design is such that it minimises chances of injury.

The company started with vehicles making trips in the neighbourhood for grocery delivery. Customers can place delivery orders on the Web or through a mobile app. Recently, the company began working with CVS Pharmacy to provide prescriptions and essentials through its vehicles in the US. The customers can place orders on the CVS website or via mobile app and select free autonomous delivery options at their preferred location during the pilot. When the autonomous Prius vehicles (and later on delivery bot R2) reach the location, users need to confirm their identity to unlock their delivery.

*—Ayushee Sharma*

# 4 SOLVING WATER CRISIS WITH AIR AND NANOTECHNOLOGY

The scarcity of clean water and its access is affecting the entire human population, be it directly or indirectly. Along with natural causes like droughts, reasons like the rising population and global warming are hugely responsible for adding to this pressure. Clean, fresh, and safe water is not only essential for drinking but also for food production, besides other important uses. Indian cities, too, will run out of or contaminate groundwater and surface water resources completely in a few years, if proper measures are not taken.

This is where an IIT Madras-incubated company VayuJal Technologies, which aims to translate advanced technologies for social good came up with a unique solution to deliver clean water suitable for drinking and other day-to-day purposes on-demand, that is, when and wherever required with quality assurance. The team, consisting of founders Thalappil Pradeep, Ramesh Kumar, and Ankit Nagar in 2017, raised a seed fund from Engineers India Limited (EIL) under the Startup Fund scheme.

Their environmentally sustainable device, atmospheric water generator (AWG) unit, captures moisture available in ambient air. The units are made with patented surface engineering technology, energy-efficient unit design, and mineralisation technology that expands their water collecting efficiency. The moisture is condensed and collected into a tank. The condenser is made using nanotechnology incorporated products such as biomimetic, nano-engineered, and micro-nano structure patterned surfaces. Modular water purifier cartridges remove undesirable impurities present in water. Re-mineralisation cartridges then add essential minerals (calcium, magnesium, iron, copper, potassium, sodium, selenium, zinc, cobalt, vanadium, and molybdenum) to purified water in a sustainable and controlled manner.

The cost is significantly less than that of currently available bottled mineral water. As these units can run on off-grid solar panel systems, these are highly beneficial in remote or rural areas where water cannot be made available due to issues like financial constraints. Energy consumption of solar products is low, which makes them suitable for commercial purposes, thereby boosting the renewable energy sector. The water quality complies with BIS - IS 2012 and all necessary World Health Organization (WHO) standards, and maintenance is